SOLAR FILAMENT ERUPTION & SATELLITE RADIATION ALERT

OFFICIAL SPACE ENVIRONMENT TELEMETRY BULLETIN | ISSUED: 2026-08-21 21:49 UTC

1. EXECUTIVE EVENT SUMMARY

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SOURCE REGION     : S4.0° E3.1° (Stonyhurst)
ERUPTION RISK     : 24h: 2.3% | 48h: 4.6% [QUIESCENT / LOW RISK (Stable)]
PROJECTED FLARE   : Sub-flaring / Quiescent Background Soft X-Ray Class
EARTH CONNECTIVITY: Magnetic Separation Δφ = 1.9° (High-Flux Parker Spiral Path)
ASTRONAUT SAFETY  : STANDARD SPACE ENVIRONMENT PROCEDURES MAINTAINED
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2. DETECTED FILAMENT MORPHOMETRIC MEASUREMENTS

- True Physical Spine Length : 72,500 km (0.104 R_Sun)
- True Plasma Surface Area : 6.26e+08 km²
- Magnetic Neutral Line Shear: 45.0° (High Free Energy Polarity Inversion Line)
- Mean Chromospheric Contrast: 0.000 ΔI/I_0

3. CORONAL MASS EJECTION (CME) DRAG-BASED IMPACT FORECAST

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CME LAUNCH SPEED  : 550.0 km/s (Estimated Initial Plasma Ejection)
1-AU ARRIVAL SPEED: 479.6 km/s (Aerodynamically Decelerated)
TRANSIT TIME      : 81.9 Hours (3.4 Days)
ESTIMATED IMPACT  : 2026-08-25 07:42 UTC
GEOMAGNETIC SCALE : NOAA G1 (Minor) | Kp Index: 5.0/9
POWER GRID IMPACT : Intermittent voltage regulation alarms; induced pipeline currents
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4. OPERATIONAL AEROSPACE MISSION DIRECTIVES

- Human Spaceflight (ISS / Tiangong) : Restrict crew to radiation-hardened service modules.
- Lagrange Sentinels (DSCOVR / JWST) : Throttle optical sensors; enable radiation scrubbing on SSR.
- GNSS Navigation (GPS III / Galileo): Upload clock phase corrections; monitor L-band scintillation.
- LEO Mega-Constellations (Starlink) : Orient satellites edge-on (knife-edge) to mitigate drag decay.